Xu-Cheng Wang

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Education

Fudan University, Department of Physics Shanghai, China
PhD in Theoretical Physics (Condensed Matter Theory) *Sept. 2021 – Present*

– *Elite PhD scholarship, GPA: 3.83/4.0*

– *Supervised by Prof. Yang Qi*

– *Thesis title: Preformed pairs and phase fluctuations in two-dimensional superconductors*

Fudan University, Department of Physics Shanghai, China
B.S. in Physics, GPA: 3.65/4.0 (top 15%) *Sept. 2017 – Jun. 2021*

Academic Research

Phase fluctuations, charge and magnetic pseudogap in 2D superconductors

- We theoretically studied the phase fluctuations in generic two-dimensional superconductors and predicted the emergence of charge pseudogap, Fermi arcs, and magnetic pseudogap without invoking competing order.
- We validated these predictions by simulating strongly correlated models using determinant quantum Monte Carlo. The associated DQMC and SAC implementations are available as open-source code at `dqmc-framework` and `sac`.
- **Related papers**:
  1. **X.-C. Wang** and Yang Qi, *Phase fluctuations in two-dimensional superconductors and pseudogap phenomenon*, Phys. Rev. B, **107**, 224502 (2023).
  2. **X.-C. Wang**, Xiao Yan Xu, and Yang Qi, *The interplay of phase fluctuations and nodal quasiparticles: ubiquitous Fermi arcs in two-dimensional d-wave superconductors*, Sci. Bull. **71**, 36-39 (2026).
  3. **X.-C. Wang** and Yang Qi, *Emergent magnetic pseudogap from phase fluctuations and hierarchy of scales in two-dimensional superconductors*, arXiv:2608.07238 (2026).

Quantum geometric localization length in an ideally flat Chern band

- We showed that the localization length in an isolated, ideally flat Chern band is a quantum geometric length. The quantum geometry also governs the criticality of localization transition and drives a flow of critical exponents connecting the Dirac and orthogonal fixed points.
- Our findings provide a novel quantum geometric perspective on the localization in quantum Hall systems and shed new light on the long-standing controversy over the criticality of the integer quantum Hall transition.
- **X.-C. Wang** and Yang Qi, *Quantum geometric localization length and localization criticality in an ideally flat Chern band*, arXiv:2608.08042 (2026).

Nullspace-guided adaptive bootstrap of quantum many-body systems

- The many-body bootstrap is a relaxation approach that yields rigorous lower bounds on ground-state energies by optimizing over an outer approximation to the physical feasible set. We developed the nullspace-guided adaptive bootstrap, which dynamically refines the operator basis using the moment matrix nullspace and iteratively improves the energy bound.
- The method yields nearly exact ground-state energy bounds for the transverse-field Ising chain and improves state-of-the-art lower bounds for the Hubbard chain by up to two orders of magnitude. It provides a practical and scalable route towards accurate certified bounds on the ground-state energies and general observables of quantum many-body systems. An open-source implementation is available at `QMBBoot-NGA`.
- **X.-C. Wang** and Yang Qi, *Nullspace-guided adaptive bootstrap of quantum many-body systems*, to appear (2026).

Languages & Technical Skills

Languages: Mandarin, English
Programming: C/C++, Python, Linux, HPC, AI Agents